# HDOC Day-49

## Question

Write a C program to read a positive integer n and pass its address to pivot_integer(). Find x such that the sum from 1 to x equals the sum from x to n (both inclusive). Print x, or -1 if none exists.

## C Program

```
#include <stdio.h>

void pivot_integer(int *n) {
    int total = (*n) * (*n + 1) / 2;
    int left = 0;

    for (int x = 1; x <= *n; x++) {
        left += x;
        int right = total - left + x;
        if (left == right) {
            printf("%d\n", x);
            return;
        }
    }
    printf("-1\n");
}

int main() {
    int n;
    scanf("%d", &n);
    if (n <= 0) return 0;
    pivot_integer(&n);
    return 0;
}
```

## Sample Input and Output

```
Input:
8

Output:
6
```

## Explanation

For n = 8, the sum from 1 to 6 is 21 and the sum from 6 to 8 is also 21. The function receives n through its pointer.